

ECO-PULSE

Working Proof of Concept

AESH Sustainability Hackathon 2026 · Challenge 2: Green Radar Systems · Egypt, July 2026



1. Proof of Concept Type

Fully executable **software simulation and validated analytical model** developed natively in MATLAB/GNU Octave. The model requires no proprietary toolboxes or physical hardware to robustly demonstrate its core functionalities.



2. Access to the Working Model

Project Repository:

<https://github.com/bemwahany145-bit/ECO-PULSE.git>

(Replace the URL above with your actual repository link before exporting)



3. Demonstration of Core Functions

To execute the simulation and verify the core capabilities (adaptive duty-cycling, real-time tracking, and flood forecasting), judges can run the following primary scripts directly from the repository root:



`radar_gui.m`

Launches the interactive 7-tab GUI, demonstrating the live dashboard, 3D digital twin, and real-time AI decision-making.



`compare_baseline_vs_green.m`

Executes the comparative simulation across 5 distinct environmental scenarios (*stable*, *slow_rise*, *fast_rise*, *receding*, *flash_flood*) to prove the dynamic energy-saving capabilities of the radar.



`run_full_validation.m`

Runs the full validation suite, outputting the primary technical (CFAR optimization, forecast RMSE) and sustainability KPIs directly to the console.

 All scripts generate automated logs and visualizations to validate the reported technical claims.